

Cybercrime Case Study: Phishing

The Crime

Phishing is a form of social engineering in which an attacker impersonates a trusted entity – a bank, a company, or an institution – to trick a victim into surrendering sensitive information such as login credentials, OTPs, or financial details. Unlike brute-force hacking, phishing exploits human psychology rather than technical vulnerabilities. It is the most commonly reported cybercrime in India, accounting for a significant portion of cyber fraud complaints filed on the National Cyber Crime Reporting Portal (cybercrime.gov.in).

How It Typically Happens – Step by Step

Step 1 – Target Selection: The attacker identifies a target group. Students actively searching for internships, jobs, or scholarships are a common category, as they are likely to respond quickly to unsolicited "opportunities."

Step 2 – Crafting the Lure: The attacker creates a convincing email or message. It includes a recognisable brand logo, professional formatting, and urgent language ("Limited slots available – respond within 12 hours").

Step 3 – The Hook: The message contains a link to a fake website that mirrors a legitimate one. The URL is slightly altered – for example, *techcorp-apply.in* instead of *techcorp.in* – a detail most users do not notice.

Step 4 – Credential Harvesting: The victim enters their name, contact details, email address, and in many cases, their login credentials – often framed as "verification" by the fake form.

Step 5 – Exploitation: The attacker uses harvested credentials to access accounts, extract personal data, commit financial fraud, or sell the information on dark web marketplaces.

Who Is Targeted

Phishing attacks disproportionately target three groups: students and young professionals (due to urgency around opportunities and

relatively lower awareness), senior citizens (unfamiliar with digital fraud patterns), and employees in organisations with access to financial systems or confidential data. In academic contexts, students applying for internships, scholarships, or competitive programmes are especially vulnerable – they are motivated, in a time-pressured mindset, and often encounter legitimate opportunity emails daily, making fraudulent ones harder to distinguish.

Consequences

The consequences of phishing operate at multiple levels.

For the individual: Immediate financial loss if banking credentials are compromised, unauthorised access to email and social media accounts, identity theft through misuse of personal documents, and long-term psychological distress – anxiety, loss of trust in digital systems, and a sense of violation.

For institutions: If an attacker gains access through a student or employee's credentials, the damage can extend to institutional data, examinations systems, or research databases.

In documented cases: The 2020 Twitter phishing attack – in which attackers used social engineering to manipulate Twitter employees into surrendering internal access credentials – resulted in the compromise of over 130 high-profile accounts and the theft of approximately USD 120,000 in Bitcoin. (*Source: U.S. Department of Justice, Case No. 20-cr-00452, 2020.*)

What Could Have Been Done Differently

In most phishing incidents, a small number of deliberate actions would have prevented compromise entirely: verifying the sender's email address carefully, accessing opportunities only through official websites rather than email links, and treating any request for login credentials as an automatic red flag – since no legitimate organisation requests passwords via a form. The challenge, as Arjun's case in this report illustrates, is not the absence of this knowledge – it is the failure to apply it under pressure.
